INCH-POUND
MIL-DTL-24640/12E
8 November 2011
SUPERSEDING
MIL-DTL-24640/12D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 26 AWG, TYPES 2XO AND 2XOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (2XO) and Watertight (2XOW)

First	-	19-strand, tin-coated copper conductor, 26 AWG. Watertight constructions shall be constructed with 7 strands.
Second	-	Component wires shall be SAE-AS81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed in the specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Two conductors twisted together. Pair lay length shall be in accordance with table II. Twisted pair identification code, one black and one white, applied using Method 6, or telephone color code applied using Method 3. For 2XO-77, if telephone color code is used, repeat the color code of pairs 1-11, in sequence, for pairs 67-77.
Fourth	-	The required number of twisted pairs shall be cabled together in accordance with table III with a lay length not greater than 16 times the diameter over the outermost layer. Cable sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used, as necessary, to form a round cable.
Fifth	-	Optional binder tape.
Sixth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Seventh	-	Optional binder tape.
Eighth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

PIN M24640/12	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UO	2XO-6	6	0.038	0.305	0.329	81.6
-02UO	2XO-18	18	0.041	0.417	0.449	140.0
-03UO	2XO-24	24	0.044	0.473	0.509	175.0
-04UO	2XO-42	42	0.044	0.565	0.609	275.0
-05UO	2XO-60	60	0.047	0.641	0.691	334.0
-06UO	2XO-77	77	0.047	0.728	0.785	395.0
-07UO	2XOW-6	6	0.038	0.310	0.340	78.0
-08UO	2XOW-18	18	0.041	0.420	0.460	129.0
-09UO	2XOW-24	24	0.044	0.475	0.525	178.0
-10UO	2XOW-42	42	0.044	0.570	0.630	243.0
-11UO	2XOW-60	60	0.047	0.660	0.710	315.0
-12UO	2XOW-77	77	0.047	0.740	0.800	400.0

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TABLE II. Pair lay [1/].

Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)
1	1.5	27	1.0	53	2.5
2	1.0	28	2.0	54	3.0
3	2.0	29	3.0	55	4.5
4	3.0	30	4.5	56	5.5
5	4.5	31	1.0	57	1.0
6	1.0	32	2.0	58	1.5
7	2.0	33	3.0	59	2.0
8	3.0	34	4.5	60	2.5
9	1.5	35	1.0	61	3.0
10	2.5	36	2.0	62	4.5
11	5.5	37	3.0	63	5.5
12	1.5	38	4.5	64	1.0
13	2.5	39	1.0	65	1.5
14	3.7	40	2.0	66	2.0
15	5.5	41	3.0	67	2.5
16	1.5	42	4.5	68	3.0
17	2.5	43	1.0	69	4.5
18	3.7	44	1.5	70	5.5
19	1.5	45	2.0	71	1.0
20	2.5	46	2.5	72	1.5
21	3.7	47	3.0	73	2.0
22	5.5	48	4.5	74	2.5
23	1.0	49	5.5	75	3.0
24	2.0	50	1.0	76	4.5
25	3.0	51	1.5	77	5.5
26	4.5	52	2.0		

NOTE:
[1/] All lays are left-handed.

TABLE III. Assembly.

Type number	Core	Assembly			
		Layer 1	Layer 2	Layer 3	Layer 4
2XO/W-6	---	Pairs 1-6	---	---	---
2XO/W-18	Pair 1	Pairs 2-6	Pairs 7-18	---	---
2XO/W-24	Pairs 1-2	Pairs 3-10	Pairs 11-24	---	---
2XO/W-42	---	Pairs 1-8	Pairs 9-22	Pairs 23-42	---
2XO/W-60	Pair 1	Pairs 2-7	Pairs 8-19	Pairs 20-36	Pairs 37-60
2XO/W-77	Pairs 1-3	Pairs 4-12	Pairs 13-27	Pairs 28-49	Pairs 50-77

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REQUIREMENTS:

Qualification required.

Construction details: See tables I, II, and III.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	
Non-blocked	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Water-blocked	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test	
Primary insulation (when tested in lieu of spark test)	6.0 kilovolts (peak), 100 percent impulse test
Finished wire	8.0 kilovolts (peak), 3 kHz spark test 5.5 kV, 100 percent test
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	
Non-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 168 hours
Water-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours
Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)
Shrinkage	
Non-blocked	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Water-blocked	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Spark test primary insulation	1500 volts (root mean square), 3 kHz, 100 percent test
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch

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Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	44.24
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Component shield to shield	Not applicable
Insulation resistance (megohms/1000 feet, min.)	2500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot(pF/ft), max.)	33
Capacitance unbalance (percent, max.)	15
Characteristic impedance (at 1 MHz, ohms)	75 (min.), 105 (max.)
Watertightness (type 2XOW only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Drip (type 2XOW only)	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable

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Group C:

Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	25
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (type 2XOW only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10

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Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
All types and sizes 6 - 24	500
All types and sizes 42 - 77	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-080)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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